Scientists are one step closer to 3D-print a full-sized, adult human heart

A team of researchers from Carnegie Mellon University has published a paper that details a technique that allows anyone to 3D-bioprint tissue scaffolds out of collagen, the major structural protein in the human body. This first-of-its-kind method brings the field of tissue engineering one step closer to being able to 3D-print a full-sized, adult human heart.

The technique, known as Freeform Reversible Embedding of Suspended Hydrogels (FRESH), has allowed researchers to overcome many challenges associated with existing 3D-bioprinting methods, and to achieve unprecedented resolution and fidelity using soft and living materials.

Each organ in the human body, such as the heart, is built from specialised cells that are held together by a biological scaffold called extracellular matrix (ECM). This network of ECM proteins provides the structure and biochemical signals that cells need to carry out their normal function. However, until now it has not been possible to rebuild this complex ECM architecture using traditional bio-fabrication methods.

Adam Feinberg, a professor of biomedical engineering, and materials science and engineering at Carnegie Mellon, whose lab performed this work, says, "What we have shown is that we can print pieces of the heart out of cells and collagen into parts that truly function, like a heart valve or a small beating ventricle. By using MRI data of a human heart, we were able to accurately reproduce patient-specific anatomical structure and 3D-bioprint collagen and human heart cells."

Scientist creates whiskers for drones

A University of Queensland engineer has developed whiskers for drones and robots, allowing them to sense surroundings just as animals do. Dr Pauline Pounds says that her team wanted to build affordable sensors to help robots work with peri-personal space—the region immediately around but not touching a person. Dr Pound adds, "Whether it is a humanoid teaching robot in the classroom or hovering drones in flight, being able to sense interactions before contact is important."

The whiskers are extremely sensitive and can detect minute forces such as from motion of air to be able to measure human breath from half a metre away. This allows smaller drones to navigate and stabilise flight through dark, dusty, smoky, cramped spaces or gusty, turbulent environments without having to mount heavier sensors.

"The whiskers are long slender fibre hairs made from the same plastic material that 3D printer extruders use. These are attached to small force-transmitting plates that are glued onto a miniature tripod of pressure sensors, which can then detect tip loads as low as 0.33 milligrams, which is less than the weight of a flea," she explains.

The first application of the whiskers was on a robot rat called iRat, used to help the study of rodent psychology and neurology (Credit: University of Queensland)

Dr Pound explains, "The whiskers can be used to measure fluid velocity, as well as to detect the bow-wave of oncoming air of an approaching object before it actually touches the whiskers. These can be carried anywhere to measure force, like in machining applications, in industrial fabrication, in medicine, in marine systems, in aerospace—the possibilities are endless."

Norm are the next-generation smart glasses

Launched at CES 2019, Norm glasses, by Human Capable Inc., look almost like regular sunglasses, and yet these have a minicomputer embedded inside the frame, including CPU, memory, battery, speakers, microphones, camera and a display system that can project digital information into the user's field-of-view.

Norm is a fully-functional mini-computer (Credit: Kickstarter)

With an Android-based system, Norm's functionalities can be extended with apps in the same way as on smartphones. In addition, Norm supports integration with voice platforms such as Alexa—functionalities are available to the user anywhere he or she goes.

With excellent image quality, unobtrusive eyebox and twenty-degree field-of-view, the user can enjoy digital content with ease, even under bright sunshine. He or she can interact with Norm using voice or gesture control. Norm uses offline voice recognition that works regardless of an Internet connection. In addition, a touch sensor on the right temple allows for simple gesture controls to navigate the interface.

Norm pairs via Bluetooth with any smartphone. The user can answer calls, read or reply to texts, without touching the phone. Norm can be personalised to match personal style, needs or preferences, too.